

CI/CD Pipeline — Automated Lambda Deployment

Every GitHub push deploys Lambda automatically — zero manual steps ap-south-1



Developer
git push

git push origin main



GitHub
aws-cicd-pipeline

 Webhook fires within 60 seconds



CodePipeline
AWS CodePipeline

Stage 1 — Source

Detects GitHub push via webhook. Downloads source code from the main branch and stores it in the S3 artifact bucket. Orchestrates the entire pipeline.

Source Stage

GitHub v1 Provider

OAuthToken Auth

output: source_output



IAM Roles
3 Roles — Least Privilege

 pipeline-role → S3 + CodeBuild

 build-role → S3 + CW + Lambda

 lambda-role → Basic Execution



CloudWatch
Build Logs + Monitoring

/aws/codebuild/aws-cicd-pipeline

Every echo streamed live

Build duration tracked



S3 Artifact
Amazon S3

Stores build artifacts between pipeline stages. Code zip passed from Source → Build.

Versioning ON

force_destroy



CodeBuild
AWS CodeBuild

Stage 2 — Build

Runs buildspec.yml — installs Python 3.11, zips Lambda code, deploys via AWS CLI.

standard:7.0

BUILD_GENERAL1_SMALL



Terraform IaC
All infrastructure as code

s3.tf · iam.tf · lambda.tf

codebuild.tf · codepipeline.tf

terraform apply → 9 resources

 buildspec.yml phases run



buildspec.yml — 4 Phases
CodeBuild Instructions

INSTALL

PRE_BUILD

runtime: python 3.11

cd lambda/ → verify files

BUILD

zip lambda_package.zip

POST_BUILD

aws lambda update-function-code

 update-function-code called



Lambda Function

AWS Lambda — aws-cicd-pipeline-demo

✓ Deployed

Updated automatically with new code. Environment variables set with build number and timestamp. New version live within 90 seconds of git push.

Python 3.11

timeout: 30s

APP_VERSION auto-updated

DEPLOYED_AT timestamp

lifecycle ignore_changes

LEGEND

Main pipeline flow

Log streaming

☐ Compute / Orchestration

☐ Deployed / Monitoring

☐ Pipeline trigger

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CodePipeline

CodeBuild

Lambda

S3

IAM

CloudWatch

Terraform